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Ligament tissue remodeling on quantitative ultrasound.

eCTM ligaments reached a responding Tissue Remodeling Score by Week 12; standard of care did not.

Background

OBJECTIVE

Soft-tissue injuries in performance horses are notoriously difficult to monitor. Standard B-mode ultrasound reads out qualitatively and depends on the sonographer. We asked whether a quantitative ultrasound texture score — the Tissue Remodeling Score (TRS) — could detect early, objective structural recovery after a single injection of cryopreserved umbilical allograft (equicenta® CTM) versus clinician-selected standard of care.

Methods

IMAGING & SCORING

B-mode cine loops were acquired with a Butterfly iQ3 handheld ultrasound at Weeks 0, 2, 4 and 12. ROIs were segmented on the lesion and a matched contralateral control. Six texture features were combined into a composite TRS normalized so a healthy contralateral tissue scores 50 at baseline.

SAMPLE (ITT, QC-PASSED)

46 injured tissues passed image-quality QC: 22 eCTM vs. 11 standard-of-care ligaments (primary comparison), plus 13 tendons as an underpowered exploratory cohort. Segmentation was blinded.

VALIDATION

Segmentation reliability across 25 ROIs / 8 cases: Dice 0.931 (0.843–0.947). Week-2 TRS correlated with final Week-12 TRS at $r=0.88$, and the ligament TRS trajectory correlated with 26-week performance at Cohen's $d=0.96$.

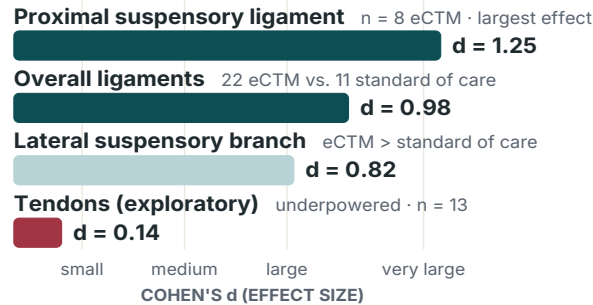
Results

LIGAMENTS REACHING THE RESPONDING TRS THRESHOLD BY WEEK 12

36% : 0%

8 / 22 eCTM vs. 0 / 11 standard of care
 $p=0.013$ · Cohen's $d=0.98$

FIG 1 · Effect sizes by tissue subgroup (Week 12)



TRS trajectory effect sizes (Cohen's d) at Week 12. Proximal suspensory ligaments show the largest structural response; tendons shown for reference were underpowered.

Interpretation

STRUCTURE AND FUNCTION CONVERGE

eCTM ligaments built responding TRS scores (at or above 55) over Weeks 2–12 while standard-of-care ligaments trended below the Not-Responding threshold (<45). Tendon TRS correlated with lameness at Weeks 2 and 4 ($r=-0.74$, $p<0.02$), and the ligament TRS trajectory correlated with 26-week clinical performance (Cohen's $d=0.96$) — an independent structural corroboration of the functional results reported in the parent trial.

Structural remodeling, not just symptom relief.

Quantitative ultrasound detected an early, blinded, tissue-level signal of recovery in eCTM ligaments that clinician-selected standard of care did not produce.

Equine Performance Labs